# Andy Liu

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## Education

**University of Toronto** Sept 2024 – Apr 2029 (Expected)

*Bachelor of Applied Science in Engineering Science, Major in Electrical & Computer Engineering, Minor in Robotics & AI*

- Coursework: Electronics (Op-Amps, Filters, Digital Systems, Circuit Analysis, Instrumentation), Digital Systems (SystemVerilog, Logic Gates), Computer Organization (FPGA, RISC-V Assembly)

**University of Waterloo** Sep 2023 – Aug 2024

*Bachelor of Math in Mathematics/Financial Analysis and Risk Management*

- Coursework: Algorithms & Data Structures (C), Tools for Software Dev. (Bash, Git, CMake), Public Speaking

## Skills

**Embedded Systems:** Altium Designer, KiCad, Soldering, Oscilloscopes, Multimeters, Power Supplies, ESP32, Arduino
**Languages:** Python, C, C++, SystemVerilog, RISC-V Assembly, Java, JavaScript, HTML & CSS
**Engineering Tools:** Git & GitHub, Scripting (Bash), CMake, CAD (Onshape)

## Projects

**Wannabe WALL-E - 1st Place Autonomous Vehicle Challenge** Feb 2025

- Won 1st place in the UTRAHacks 2025 Autonomous Vehicle Challenge, outperforming 250+ participants.
- Built an **Arduino** autonomous robot that navigated a multi-stage obstacle course using ultrasonic and IR sensors.
- Engineered a corrective turning algorithm to compensate for a malfunctioning motor driver, averting critical failure.

**Resonant Evil - Audio-Based Hardware Maze Escape Arcade Game - Hackathon Winner** Feb 2025

- Won the Video Games Theme Prize at MakeUofT 2025, Canada's largest hardware hackathon (300+ participants).
- Developed an immersive audio-driven **Arduino** arcade game designed to increase accessibility for the visually impaired.
- Designed dynamic audio and haptic feedback system to guide player navigation and build tension.

**Silent Night - IoT Smart Pillow - Hackathon Winner** Feb 2024

- Won the Smart Home Tech Theme Prize at MakeUofT 2024, Canada's largest hardware hackathon (300+ participants).
- Designed an IoT smart pillow using an **ESP32** and sensor array to monitor sleep conditions in real-time.
- Built an automated posture-correction system micro-adjustments to improve sleep ergonomics.

**BreadCrumb - Deep Learning Project** Feb 2026 – Mar 2026

- Built a **PyTorch** object-detection model to classify and count breadboard components via bounding boxes and NMS.
- Collected and labeled a custom circuit-image dataset in **Roboflow** to support supervised learning.
- Fine-tuned a pretrained ResNet18, iterating on architecture and hyperparameters to reach 97%+ accuracy.

## Experiences

**Tech Team Director** *(Previously Project Lead)* Toronto, ON

*IEEE University of Toronto Branch* Aug 2025 – Present

- Devised the Smart Cane project pipeline, linking object detection & sensor readings to dynamic haptic feedback.
- Coordinated cross-functional contributions of a 5-person team on the Smart Cane Project, ensuring steady progress.
- Overhauled documentation for IEEE's technical workshop series (300+ students/year) to eliminate redundant prep work.
- Ran a **Git & GitHub** workshop for 100+ hackathon participants, teaching version control fundamentals hands-on.
- Led an **Altium** workshop for 30+ participants, teaching foundational PCB schematic design principles.

**Embedded Systems & Machine Learning Research Intern** Markham, ON

*CAROBOT Learning and Research Organization* May 2026 – Sept 2026, May 2025 – Aug 2025

- Enabled fully offline, browser-based access to CheeseBlocks, a block-based embedded programming tool.
- Built CheeseBlocks' data-to-inference ML pipeline: in-browser training (**TensorFlow.js**), on-device weight deployment.
- Streamlined ESP32-S3 firmware setup via a web flasher, bundling **MicroPython**, ulab, and camera support.